# Data Management Plan (DMP) -

| | | |
|---|---|---|
| | **Introduction and general descriptions** | This DMP generally refers to all data processed in the research project "Newcomers in high mountain areas of the Austrian Alps" which is entirely funded by the Austrian Science Fund (P 32956). Based on the project's progress and the data collected during this period (2020/09/01-2023/08/31), the DMP is continuously updated. Therefore, it should be seen as an living document with open output.<br>Depending on the legal and ethical restrictions (c.f. Chapter IV) of the managed data, selected parts of it are also archived long-term in a repository. Currently the research data will be deposited at the Austrian Social Science Data Archive (AUSSDA). |
| | **Data Officer and further contact** | E. Steinicke (Ernst.Steinicke@uibk.ac.at) is responsible for the DMP as project initiator (PI), while M. Hartlieb (Martin.Hartlieb@uibk.ac.at), IT technician at the institute of geography (University of Innsbruck), is responsible for IT-specific questions (server infrastructure, storage volume, backups, encoding e.g.). B. Grüner (Bernhard.Gruener@uibk.ac.at) is responsible for the coordination and communication of long-term archiving (e.g., project-related research data and publications) on the selected repositories.<br>The contact person at AUSSDA is L. Hönegger (info@aussda.at). |
| **I** | **Data Characteristics** | |
| **I.1** | **Description of the data** | |

### I.1.1 Used data type, format, volume

The following table shows the data types used, the formats, the expected volume required and the data processing:

| Type | Format | Volume | Process |
|---|---|---|---|
| Geodata | SHP, JPG, TIFF, etc. | max. 100 gigabyte | Generated/reused |
| Audio data | WAV, MP3, etc. | max. 10 gigabyte | Generated |
| Interview transcripts (only in German) | TXT, DOCX, PDF, etc. | max. 1 gigabyte | Generated/reused (s. column below) |
| Literature | PDF, RIS, etc. | max. 8 gigabyte | Generated/reused |
| Analysis of interviews | MX18, MOD, MTR, etc. | max. 2 gigabytes | Generated |

| | | |
|---|---|---|
| Photographs | JPG, TIFF, etc. | max. 5 gigabyte | Generated/reused |
| Vector graphics | SVG, etc. | max. 2 gigabyte | Generated |
| Raster graphics | XCF, etc. | max. 5 gigabyte | Generated |
| Videos | MP4, MOV, AVI, etc. | Max. 10 gigabyte | Generated |
| Statistical data | XLSX, CSV, ODS, SAV, etc. | 1 – 2 gigabytes | Generated/reused |

The file formats outlined above have been chosen because they are well known and accepted standards and in widespread use. Therefore, the data conversion and further processing is granted.

### I.1.2 Data encoding

During all stages of the project, the researchers will use standard character encodings (e.g. ASCII or UTF-8) while generating new data or data sets.

### I.1.3 Used methods

The methods for obtaining new data are explained below:

1) Obtaining and analyzing of Statistical data:
Data on population, agriculture, infrastructure etc. are obtained from official offices like Statistik Austria, agricultural colleges etc. Subject-specific data on newcomers' education are obtained using own questionnaires. All statistical data are analyzed qualitatively and, on a selective basis, combined with quantitative data by means of triangulation.

2) Partly structured, focus-, expert-interviews, etc.:
Qualitative methods, such as partly structured and freely conducted interviews with experts etc. are carried out. Through these initial interviews, contact is established with the newcomers. Subsequently, further newcomers will be identified by using the snowball method. Every newcomer is surveyed using a partly structured interview.

3) Network analysis:
In addition, an analysis of social networks should further break down the connections between newcomers and the autochthonous people. Furthermore, social media is chosen to contact multilocal

| | | |
|---|---|---|
| | | dwelling new highlanders. As a result the research team gains an additional perspective on the networks of new highlanders who show their connections within the case study community.<br><br>4) Mapping and analyzing of land use change:<br>Temporal comparison of historical land registry maps, photos, aerial photographs and satellite images from archives with the current situation, to record spatial effects of the new highlanders. Missing image data will be generated by fly-overs with drone. Photos from the private archives of the newcomers and image material from the research team generated during their research stays will be examined. These data will be geo-referenced and compared using a Geographical Information System (GIS).<br><br>**I.1.4 Accessibility of the DMP**<br><br>The DPM is accessible to all project members and if necessary, to members of the repositories and beyond.<br><br>**I.1.5 Target group(s) of research results**<br><br>One main target group are experts of spatial demography, agricultural sociology, ecology, natural hazard research and regional development research (dissemination via scientific journals, conferences, etc.). The other target group concerns the general public and regional policy (dissemination via conferences, workshops, print and online-media, etc.). |
| **II** | **Documentation and Metadata** | |
| **II.1** | **Metadata standards** | **II.1.1 General information**<br><br>As far as meta-data standards are concerned, we will adhere to the specifications of the selected repository.<br>In general metadata contain the Austrian Science Fund (FWF) as funding source, the project number (P 32956) as well as which data cannot be made public because of ethical or legal reasons (c.f. Chapter IV).<br><br>A few suggestions on our part: |

| | | |
|---|---|---|
| | | **II.1.2 Qualitative metadata**<br><br>The metadata of the interviews conducted contain the following information:<br>interview-ID, survey period and/or date, pseudonym of participants (structure: STATE-#NUTS1-#NUTS2#NUTS3-TG-random number, e.g. AT-3-33-AM-01), target group (TG: experts, amenity migrants, new farmers, etc.), number of participants per interview, their gender and/or age (c.f. Chapter IV), name of the case study community, federal state, region, NUTS-ID (c.f. pseudonym structure), date of realization, duration, interviewer, implementation (face to face, telephone, video-conference, e-mail correspondence), logging type (audio recording, memory protocol, text file), language, transcriber, informed consent information, ethical issues, analysis methods, analysis software.<br>For the figures generated during research, metadata is also created, which contains the following information:<br>figure-ID, file name, file format, creator, creation date, editor, editing date, additional processing information (editable with/in named software/file format), sources of used data, analysis method (formulars and/or syntax), software used.<br><br>**II.1.3 Quantitative metadata**<br><br>From which institution and for which period the data were obtained, formulas and syntax used for calculation and analysis of the raw data, information about processing, used units, format of data, additional descriptions, calculation and/or analyzing software etc. |
| **II.2** | **Documentation of data** | **II.2.1 Machine-readability**<br><br>The research team uses whenever possible, open-source software. This simplifies the subsequent reading and processing of the data for second and third parties. Where open-source applications reach their limits or affect the quality of the research, commercial software will be used. Thus, machine readability and long-term storage are also guaranteed, and can be controlled by the chosen repositories. |

| | | |
|---|---|---|
| | | **II.2.2 FAIR data access**<br><br>All publications and related data (depending on ethical and legal issues; c.f. Chapter IV) receive a DOI (c.f. Chapter III.1.2). The latter are stored in a repository (c.f. Chapter III.1.1) and are provided with the required metadata (c.f. Chapter II).<br>The publications are published in journals that guarantee open access under an appropriate license (Creative Commons Attribution CC BY, etc.). The latter clearly defines the use, disclosure and processing of data (c.f. Chapter III.1.1). All journals are listed in the Directory of Open Access Journals (DOAJ). The DOAJ lists all journals that apply with the requirements mentioned above.<br>The data and metadata are archived in a repository in accordance with the FAIR principles (c.f. Chapter III.1). In the publications, there will be a reference (DOI, etc.) to the availability of project-related long-term archived data and metadata in the repositories.<br><br>**II.2.3 Final report to the FWF**<br><br>Open access activities and compliance with the Open Access policy must be demonstrated in the final report to the FWF. This is done by specifying the persistent identifiers (e.g. DOI) under which the research data can be viewed and downloaded. |
| **II.3** | **Data quality control** | **II.3.1 Quality assurance procedures**<br><br>Quantitative data are cross-checked by the project members with regard to the quality criteria objectivity, reliability and validity. The objectivity of official statistics, on the other hand, cannot be verified by the researchers but any calculations that may have been made are also cross-checked by the project members. Prior to long-term archiving, the data quality is again verified on a random basis by the employees of the repositories.<br>The inductively and deductively generated codes of text analysis (qualitative data) are crossed-checked by the project members. This is ensured by intensive training in advance and constant discussions with subsequent re-evaluation of the codes during the analysis process in order to guarantee the highest possible intercoder reliability. |

| | | |
|---|---|---|
| | | Before recording and evaluating land use changes in GIS, all project members are required to define an accurate framework for digitization (accuracy, layer properties, attributes, coordinate system, etc.).<br><br>**II.3.2 Controlling and documentation**<br><br>The data and metadata itself are documented in an associated, machine-readable file or directly in the log files, depending on the software.<br>The quality assurance methods are always documented in the respective metadata and controlled via peer-review of several project members and via the double-blind review process upon submission of publications. |
| **III** | **Data Availability and Storage** | |
| **III.1** | **Data sharing strategy** | **III.1.1 Accessibility and used repository**<br>Selected research data (c.f. chapter IV), in particular published results, as well as those data and metadata that support their understanding will be stored in appropriate repositories. Thereby only repositories are used, that are certificated (e.g. CoreTrustSeal) and fulfill the Criteria for the Selection of Trustworthy Repositories by Science Europe or similar. Additionally, they must be listed in the re3data.org database. Furthermore, storage and access should be possible free of charge. The metadata is stored in machine-readable formats (c.f. Chapter II.2.1) where possible.<br><br>**III.1.2 Persistent identifier**<br>All published data will receive a Digital Object Identifier (DOI). This enables easy retrieval, even for long-term archiving and will be used for demonstration to the FWF (c.f. Chapter II.2.3) |
| **III.2** | **Data storage strategy** | **III.2.1 Data volume**<br>The expected total size of the data will be approximately 130 gigabytes. A large part of this is due to the storage of high-resolution satellite images and aerial photographs or their processing and |

| | | |
|---|---|---|
| | | analysis in GIS. Far less is calculated for long-term archiving in the repositories, as not all data is published.<br><br>**III.2.2 Local storage**<br>The storage and backup of all project-relevant data will be initiated by the institutes' IT technician on the servers of the University of Innsbruck. To prevent data loss, one backup takes place once a day on the server of the institute for geography, a second one is stored off-site by the university of Innsbruck. This type of data storage does not generate any costs.<br><br>**III.2.3 Long-term storage, archiving and availability**<br>The long-term archiving of selected data takes place for at least 10 years in a suitable repository (c.f. Chapter III.1.1). Consequently, only data that do not raise any ethical or legal concerns can be archived or made available (c.f. Chapter IV). How specific data is long-term archived and made available is the task of the selected repository. |
| **IV** | **Legal and Ethical Aspects** | |
| **IV.1** | **Legal aspects** | **IV.1.1 Limits to unrestricted publication**<br>Quantitative primary data (generated geodata such as arial photographs, thematic maps, etc.) as well as secondary data (official statistics, satellite images, photographs, etc.) are subject to legal restrictions.<br>The use of photographs is specified in the Austrian Copyright Law (Urheberrechtsgesetz § 42, § 54-55). Accordingly, photographs of private property (houses, agricultural land, etc.) are only allowed if they are taken from publicly accessible places (streets, squares or parks). Photos taken during drone fly-overs and the geodata obtained from them therefore require explicit consent of the owners.<br>Official statistics or data that are only available on payment often do not have a corresponding license and thus, cannot be fully published in a repository.<br>Thus, the rights to publish the data outlined above must be examined individually for each case and cannot be permitted per se.<br><br>**IV.1.2 Restriction of re-use** |

| | | |
|---|---|---|
| | | Both the publication and the re-use of the above-mentioned data must be clarified on a case-by-case basis according to legal aspects: Secondary data can only be re-used under their specified license conditions. Where possible, the primary data collected in the project will be provided with a suitable license by the repository to ensure unrestricted re-use. However, this is subject to legal (c.f. Chapter IV.1.1) and ethical (c.f. Chapter IV.2.1) aspects, but we are trying to grant a CC BY-SA 3.0 AT license where possible. In this case, the archived data may be re-used under the same license conditions and the authors as well as any changes made must be cited. |
| **IV.2** | **Ethical aspects** | **IV.2.1 Limits to unrestricted publication**<br>When publishing qualitative data, especially interview content, ethical aspects must be taken into account:<br>The target group of newcomers (amenity migrants and new farmers) interviewed in this project is a particularly vulnerable minority within each case study community. In addition, all communities are very limited in terms of space and population. For these two reasons, the publishing of the interviewees personal data and metadata (e.g., age, gender, family status, education, occupation, etc.) in a repository is not possible per se. The danger of identifying individuals is too great. To prevent this, strict pseudonymisation of the data would be necessary. However, personal information (places, names, etc.) must be pseudonymised to such an extent that the interview contents can no longer be interpreted in a meaningful way. In order not to restrict the lives of the newcomers in the case study communities through our research, no interview content or sensitive personal data will be archived for ethical reasons.<br>The publication of individual interview excerpts in journals only takes place if pseudonymisation can be used sensibly and it supports a better understanding of our results.<br><br>**IV.2.2 Security policies during data storage**<br>The computers and servers of the University of Innsbruck are equipped with the necessary security features (hardware and software). All project team members are aware that they work with sensitive personal data and how to deal with it.<br>The project-related data will be irrevocably deleted after the expiry of the project runtime to ensure the protection of personal data of the interviewees on the one hand and to comply with informed consent on the other hand.<br>Quantitative personal data is passed on to the repositories pseudonymously and archived there. Only the research team can subsequently identify individuals through a pseudonym table. |

| | | |
|---|---|---|
| | | Furthermore, encrypted systems are preferred for communication and data exchange. |